

ONA II, Norway

WMO: 012120

Lat: 62.8586N

Lon: 6.5378E

Elev: 49

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	25.6	28.4	13.3	10.8	29.2	16.4	12.7	30.8	52.3	42.5	48.2	42.4	13.2	100	0.855

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.7	66.9	59.8	64.4	58.7	62.5	57.7	61.2	64.9	59.9	63.1	58.6	61.6	9.3	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.5	76.0	63.0	58.2	72.6	61.8	57.0	69.3	60.8	27.3	65.1	26.3	63.2	25.5	61.7	67.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
47.5	41.7	36.6	24.6	72.8	3.6	3.2	21.9	75.1	19.8	77.0	17.8	78.8	15.1	81.1
			21.5	63.2	3.2	2.3	19.2	64.8	17.3	66.2	15.5	67.5	13.2	69.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.6	38.5	38.1	39.1	43.0	47.3	52.0	56.6	57.7	54.6	48.5	43.3	40.1
	DBStd	8.21	4.55	4.58	4.31	4.13	4.30	3.71	3.83	3.26	3.88	4.78	5.06	5.29
	HDD50	1969	357	334	339	214	105	17	0	0	7	84	206	306
	HDD65	6713	822	754	804	660	547	390	263	227	313	510	652	771
	CDD50	736	0	0	0	4	23	78	204	240	144	39	4	0
	CDD65	3	0	0	0	0	0	0	2	1	0	0	0	0
	CDH74	3	0	0	0	0	0	0	2	1	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	16.8	20.1	18.3	18.8	15.6	14.5	14.7	13.0	13.5	17.6	17.3	17.3	20.9
	PrecAvg	48.1	4.3	3.8	3.9	3.0	2.4	3.0	3.1	3.4	5.0	5.6	4.4	5.7
(15) Precipitation	PrecMax	64.2	9.0	8.1	12.7	6.1	4.0	8.0	6.3	8.4	11.1	10.2	9.2	9.6
	PrecMin	17.5	0.0	0.0	0.0	0.6	0.3	0.0	0.7	0.0	1.5	0.7	0.0	0.6
	PrecStd	9.6	3.0	2.6	2.9	1.7	1.0	1.9	1.4	1.9	2.8	2.7	2.9	2.3
	PrecStd	9.6	3.0	2.6	2.9	1.7	1.0	1.9	1.4	1.9	2.8	2.7	2.9	2.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	49.2	48.6	49.5	59.1	63.3	67.4	72.2	70.3	65.8	61.8	55.1	51.7	48.3
	0.4% MCWB	43.5	42.5	42.7	49.5	53.1	58.5	62.7	61.6	57.2	53.1	50.1	45.4	42.0
	2% DB	47.4	46.6	47.2	53.5	59.0	62.6	67.2	66.5	63.0	58.3	52.9	49.7	46.3
	2% MCWB	42.8	41.9	42.0	46.6	51.4	56.4	60.7	60.2	56.3	52.4	48.5	45.4	42.0
	5% DB	45.9	45.2	45.8	50.3	55.6	59.3	64.3	64.0	61.1	56.1	51.4	48.4	45.4
	5% MCWB	42.3	41.2	41.5	44.9	50.1	54.5	59.3	59.4	56.1	51.1	47.0	44.6	41.6
	10% DB	44.6	44.0	44.6	48.2	53.1	56.9	62.0	62.3	59.6	54.7	50.0	47.1	44.1
	10% MCWB	41.5	40.5	41.1	44.0	48.8	53.1	58.1	58.4	55.5	50.2	46.1	43.4	40.4
	0.4% WB	45.4	45.0	44.9	50.2	54.3	59.5	63.5	63.7	59.8	54.9	51.7	48.3	45.3
	0.4% MCDB	47.3	46.7	46.7	57.1	60.5	65.1	70.0	67.5	62.2	58.0	53.6	49.7	46.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	44.3	43.6	43.6	47.5	52.2	57.2	61.5	61.6	58.4	53.6	49.7	47.0	44.0
	2% MCDB	46.2	45.2	45.8	51.7	57.6	62.1	66.0	64.7	60.8	57.0	52.0	48.8	45.8
	5% WB	43.0	42.3	42.3	46.0	50.7	55.0	59.9	60.2	57.2	52.1	48.0	45.4	42.4
	5% MCDB	45.3	44.3	44.8	49.2	55.0	58.3	63.5	63.0	59.9	55.5	50.5	47.9	44.9
	10% WB	41.7	41.1	41.3	44.7	49.2	53.5	58.5	59.0	56.0	50.8	46.5	43.6	40.6
	10% MCDB	44.2	43.5	43.9	47.4	52.6	56.3	61.8	61.6	59.0	54.3	49.7	46.5	43.5
(35) Mean Daily Temperature Range	MDBR	5.0	4.9	5.5	6.6	6.8	5.8	6.1	5.7	5.6	5.2	5.0	5.2	5.2
	5% DB MCDBR	5.9	5.6	6.3	9.9	10.8	9.6	10.1	8.3	6.9	6.7	5.8	6.9	6.9
	5% DB MCWBR	5.4	4.6	4.5	5.4	5.6	5.5	5.3	4.8	4.4	4.8	4.9	5.7	5.7
	5% WB MCDBR	5.5	5.5	5.9	8.7	9.9	8.5	9.5	7.1	6.1	6.4	5.8	6.5	6.5
	5% WB MCWBR	5.4	5.0	5.0	5.3	5.5	5.3	5.4	4.8	4.5	5.0	5.8	6.2	6.2
	5% WB MCWBR	5.4	5.0	5.0	5.3	5.5	5.3	5.4	4.8	4.5	5.0	5.8	6.2	6.2
(40) Clear-Sky Solar Irradiance	taub	0.264	0.317	0.340	0.361	0.356	0.356	0.366	0.359	0.344	0.318	0.387	0.361	0.361
	taud	2.083	2.256	2.329	2.367	2.443	2.479	2.477	2.496	2.513	2.417	2.664	2.464	2.464
	Ebn at Noon	115	196	240	261	273	275	269	260	242	199	114	61	61
	Edh at Noon	10	19	26	31	31	30	30	27	22	16	10	6	6
(44) All-Sky Solar Radiation	RadAvg	45	192	541	1058	1462	1564	1455	1116	654	293	74	25	25
	RadStd	8	39	86	113	84	158	125	106	88	45	18	3	3

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)		+0.90	+1.22	N/A	N/A	N/A	N/A	-270	-394	+113	N/A

Nomenclature: See separate page